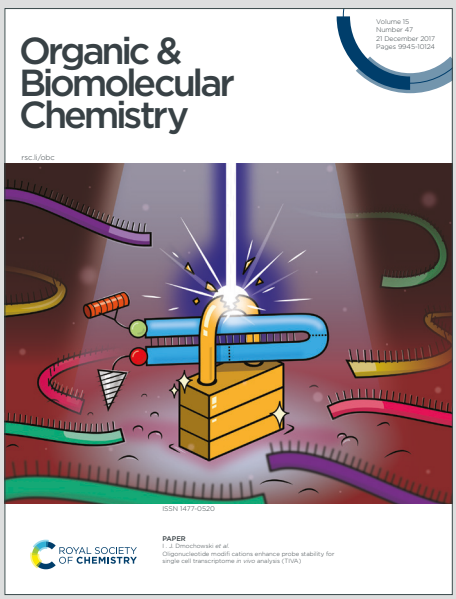
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COMMUNICATION

Palladium-Catalyzed Intermolecular Allenylation Reactions of 2,3-Disubstituted Indoles and Allenyl Carbonate

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A palladium-catalyzed intermolecular allenylation of non-strained 2,3-disubstituted indoles and allenyl carbonate has been developed, providing a convenient access to indolenines bearing an allene unit by taking advantage of the C-3 nucleophilicity of indoles. Decent yields and good functional group tolerance have been achieved with diverse indoles under mild conditions. Gram-scale reaction and various synthetic transformations have been demonstrated.

Allenes are an important class of compounds in chemical transformation[1] and natural product synthesis.[2] Functionalization of allenes is also an important means to access to allene containing various structure.[3] On the other hand, indoles are intriguing and pivotal structural motifs in numerous natural products and pharmaceuticals.[4] The studies on utilizing the nucleophilicity of C-3 of indoles to synthesize indolenines has made a great progress in recent years.[5] However, such a C3-allenylation of indoles has not been well established: the only example is the reaction of strained 1,2,3,4,5-tetrahydrocyclopta-[b]indole.[5m] Here we wish to report a general solution for the C3-allenylation of the non-strained indoles with high efficiency.

Initially, we employed 2,3-dimethyl-indole **1a**[6a] with a very low reactivity and buta-2, 3-dienyl methyl carbonate **2a**[7] as the standard substrates to explore the optimal conditions for this allenylation reaction. Fortunately, the expected product **3a** was observed in 18% yield in DCM in the presence of *t*-BuOK, Et_3B,[5e] $Pd_2(dba)_3$, and DEPhos at room temperature (Table 1, entry 1). When *t*-BuOK was replaced with NaOMe and NaH, the reaction gave poorer results (Table 1, entries 2 and 3). Increasing the dosage of *t*-BuOK did not help (Table 1, entry 4). When 2 equiv. of Et_3B were used, there was a noteworthy improvement (Table 1, entry 5). Further experimental results on solvent screening revealed that DCM and MTBE gave the best performance (Table 1, entries 5-10). Owing to better solubility of reactants in DCM, we chose it as solvent for further optimization. We then screened the different leaving groups of the allene substrates. There were remarkable improvements when OAc (**2b**), OBz (**2c**), and OBoc (**2d**) were served as leaving groups (Table 1, entries 11-13) with OBoc being the best. Employing 1.5 equiv. of **2d** could bring a slight enhancement (Table 1, entry 14). There was a decline when we continued to improve the amount of **2d** (Table 1, entry 15). With this set of optimized reaction conditions, the 1 mmol scale reaction could give a satisfactory result (Table 1, entry 16).

With the optimized reaction conditions in hand, we started to examine the scope of the reaction by applying various indoles. Different substitutions at C-5 of indole, such as F, Cl, Br, Me, MeO, are compatible, giving the corresponding allenylation products **3b**-**3f** in decent yields (Table 2, entries 1-5). Benzyl and ester groups could be tolerated at the C-3 position of indole (Table 2, entries 6 and 7).

Furthermore, we turned our attentions to using 2-non-methyl-substituted 2,3-disubstituted indoles. For 2-ethyl-3-methylindole and different non-strained hydrocycloalk[*b*]indoles, the reaction proceeded smoothly. It is worth mentioning that ketal (**1m**)[6b] is also compatible well. As a comparison, the reaction of substrate bearing a ketone unit **1n** afforded the related impure product **3n** just in 27% yield with 34% recovery of the starting material. The reaction of tetracyclic **1o** under the standard conditions also afforded the related product **3o**, which is not stable during purification by silica gel column chromatography. Thus, reduction was carried out to generate corresponding indoline *cis*-**4** (Scheme 1).[8]

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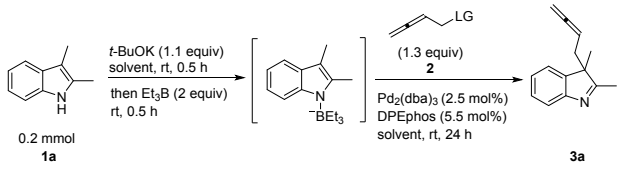
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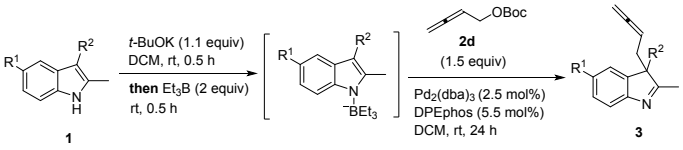
Table 1. Optimization of reaction conditions[a]



Entry	LG	Solvent	Yield of **3a** (%)[b]	NMR Recovery of **1a** (%)[b]
1[c]	OCO_2Me (**2a**)	DCM	18	69
2[c,d]	OCO_2Me (**2a**)	DCM	0	99
3[c,e]	OCO_2Me (**2a**)	DCM	6	90
4[c,f]	OCO_2Me(**2a**)	DCM	19	72
5	OCO_2Me (**2a**)	DCM	53	47
6	OCO_2Me (**2a**)	THF	6	75
7	OCO_2Me (**2a**)	Dioxane	5	79
8	OCO_2Me (**2a**)	PhMe	42	48
9	OCO_2Me (**2a**)	DMF	5	93
10	OCO_2Me (**2a**)	MTBE	51	43
11	OAc (**2b**)	DCM	80	20
12	OBz (**2c**)	DCM	80	19
13	OBoc (**2d**)	DCM	84	16
14[g]	OBoc (**2d**)	DCM	87	7
15[h]	OBoc (**2d**)	DCM	68	22
16[g,i]	OBoc (**2d**)	DCM	97 (91[j])	0

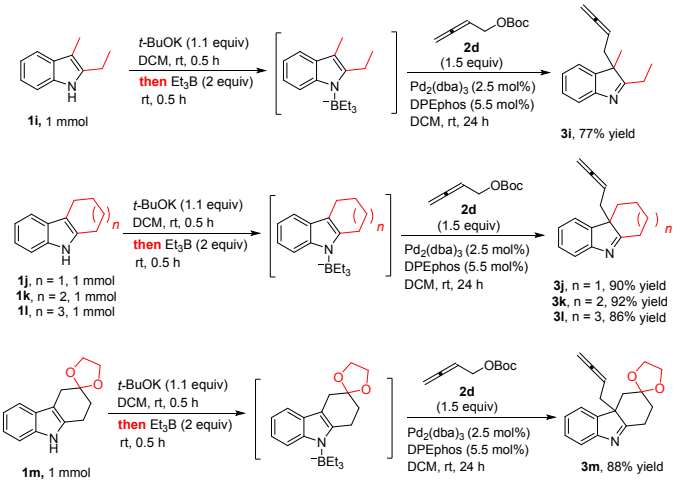
[a] Reaction conditions: The reactions were carried out using 0.2 mmol of **1a**, 1.3 equiv. of **2**, 1.1 equiv. of *t*-BuOK, 2 equiv. of Et_3B, 2.5 mol% of $Pd_2(dba)_3$, and 5.5 mol% of DPEphos in 4 mL of solvent at room temperature. [b] Determined by ^1H NMR using CH_3NO_2 as an internal standard. [c] 1 equiv. of Et_3B was added. [d] 1.1 equiv. of NaOMe was added. [e] NaH (1.1 equiv.) was added. [f] *t*-BuOK (2 equiv.) was added. [g] **2d** (1.5 equiv.) was added. [h] **2d** (1.8 equiv.) was added. [i] The reaction was carried out on 1 mmol scale in 5 mL of DCM. [j] Isolated yield.

Table 2. Substrate scope for allenylation of 2,3-disubstituted indoles with 2,3-allenyl carbonates[a]



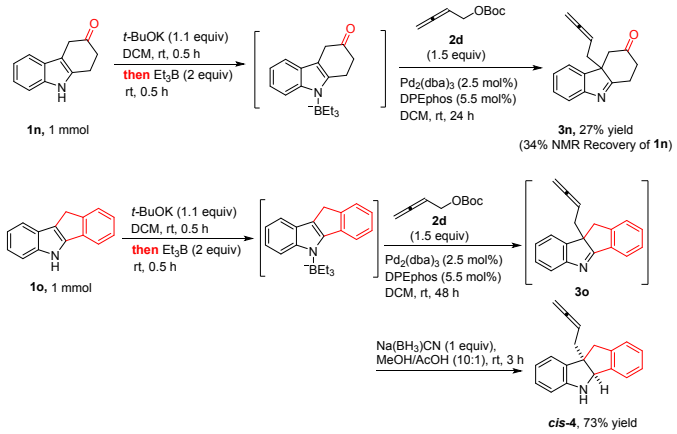
Entry	**1**		Yield of **3**[b] (%)
	R^1	R^2	
1	**F**	Me (**1b**)	93 (**3b**)
2	**Cl**	Me (**1c**)	83 (**3c**)
3	**Br**	Me (**1d**)	90 (**3d**)
4	Me	Me (**1e**)	90 (**3e**)
5	**MeO**	Me (**1f**)	80 (**3f**)
6	H	$PhCH_2$ (**1g**)	92 (**3g**)
7	H	CH_2CO_2Me (**1h**)	69 (**3h**)

[a] Reaction conditions: The reactions were carried out using 1mmol of **1**, 1.5 equiv. of **2d**, 1.1 equiv. of *t*-BuOK, 2 equiv. of Et_3B, 2.5 mol% of $Pd_2(dba)_3$, 5.5 mol % of DPEphos, and 5 mL of DCM at room temperature. [b] Isolated yield.



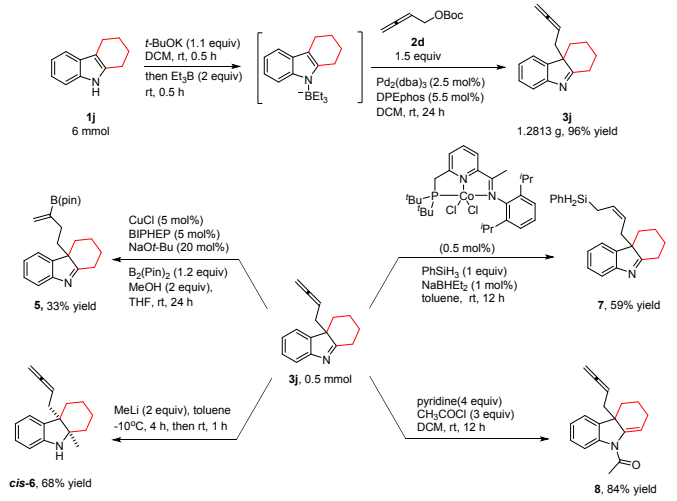
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Scheme 1. Pd-catalyzed allenylation reaction of diverse 2,3-disubstituted indoles.

The reaction was successfully conducted on 6 mmol scale of **1j**, affording **3j** in 96% yield. The copper-catalyzed borylcuprations of allenes with bis(pinacolato)diboron generated vinylboronates product **5** with a high regio- and stereoselectivity.[9] The product of *cis*-fused tetrahydrocarbazole bearing a terminal allene unit *cis*-**6** could be obtained by reacting with organolithium reagents.[10] Linear (Z)-allylsilanes **7** could be synthesized through cobalt-catalyzed allene-hydrosilylation method with a high regio- and stereoselectivity.[11] The compound **3j** could also easily react with acetyl chloride generating enamine **8** (Scheme 2).



Scheme 2. Gram-scale reaction of **1j** and synthetic application of **3j**

Conclusions

In summary, a general palladium-catalyzed intermolecular allenylation of non-strained 2,3-disubstituted indoles with 2,3-allenyl carbonate was developed affording functionalized C3-allene-substituted indolines via N−indolyltriethylborates. The reaction has exhibited good yields and a broad substrate scope under mild conditions. The products have been demonstrated to undergo some synthetically useful transformations. Further studies the reaction is ongoing in our laboratory.

Conflicts of interest

There are no conflicts to declare.

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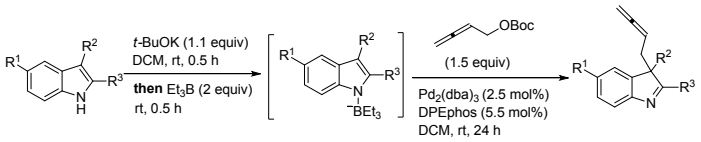
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